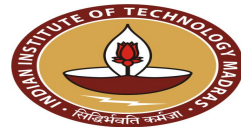


Stable Revenue Was Hiding a Silent Profit Leak

A Data-Driven Strategy to Fix Credit Leaks & Restore Margins

Mohammad Kashan

IITM BS in Data Science (BDM Capstone)



The Problem: Stable Revenue, But Critical Liquidity Risk.

Organization: Faiz General Store, Prayagraj

Model: B2C Grocery Retail (Daily Essentials)

Annual Revenue: ~₹30 Lakhs

9.45%

Current Net Margin (Critically Low)

1.86

Cash-to-Credit Ratio

Credit defaults—not demand—are the primary driver of liquidity stress.

The Audit: Converting Daily Logs into Financial Risk Signals

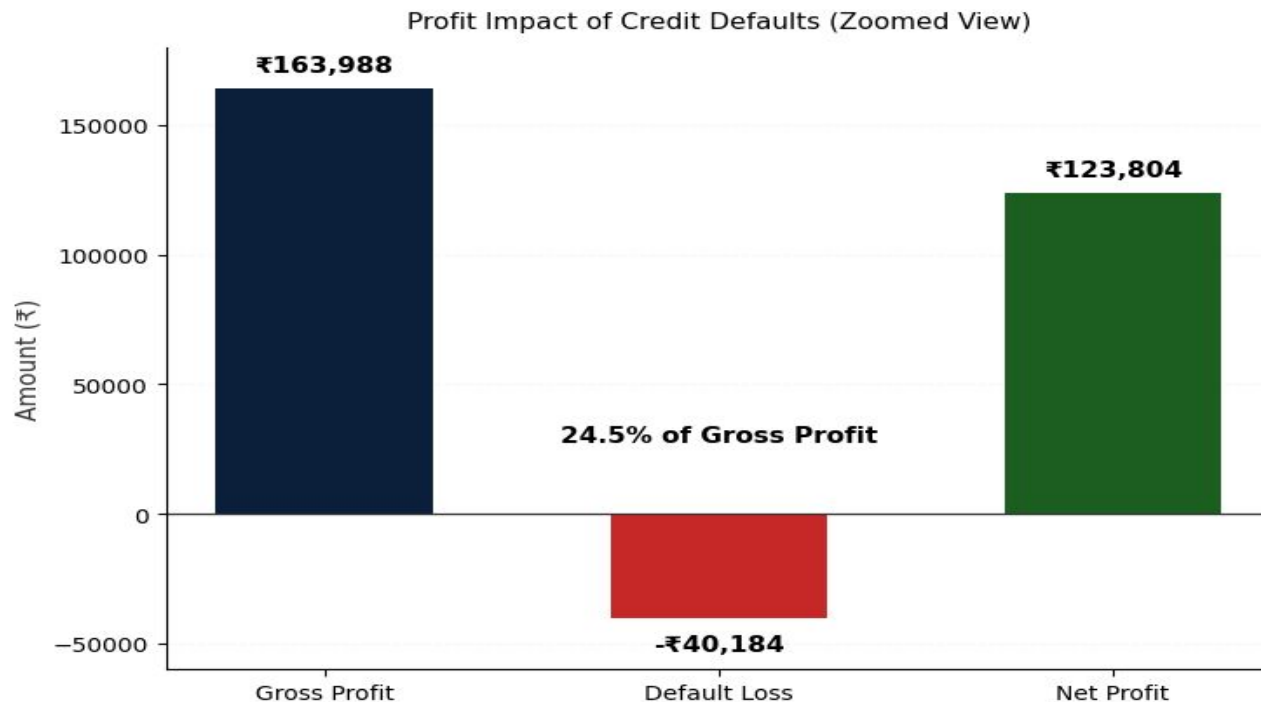
Mapped Credit Behavior: Digitized manual credit logs and daily summaries (Mar–Jul 2025) to establish customer-level credit histories.

Built a Reliability Score Model: Developed a 0–100 customer reliability score using default history and payment delays to enable tier-based credit decisions.

Stress-Tested Profit Margins: Simulated the financial impact of removing credit from high-risk customers before policy implementation.

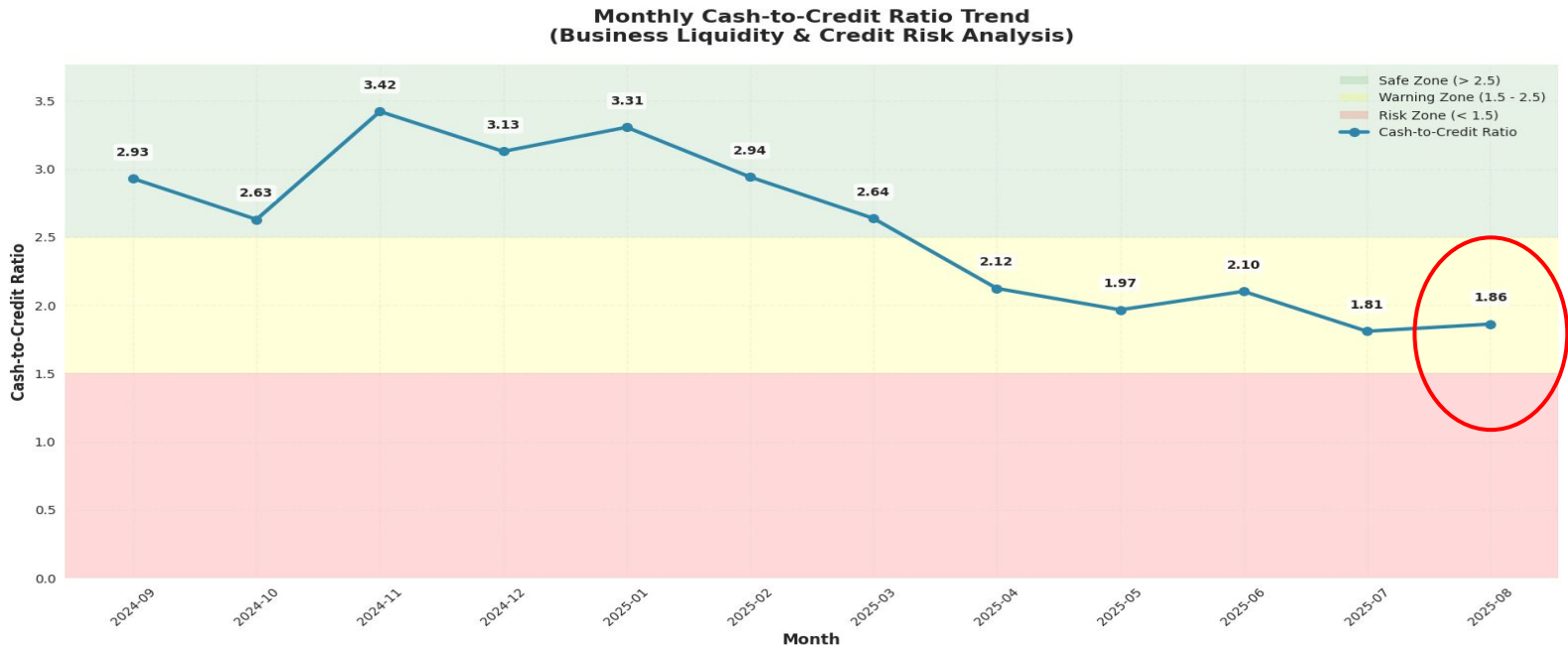
This audit transformed raw records into a deployable, policy-ready credit-risk system.

Profit Analysis: Credit Defaults Wipe Out 25% of Gross Profit



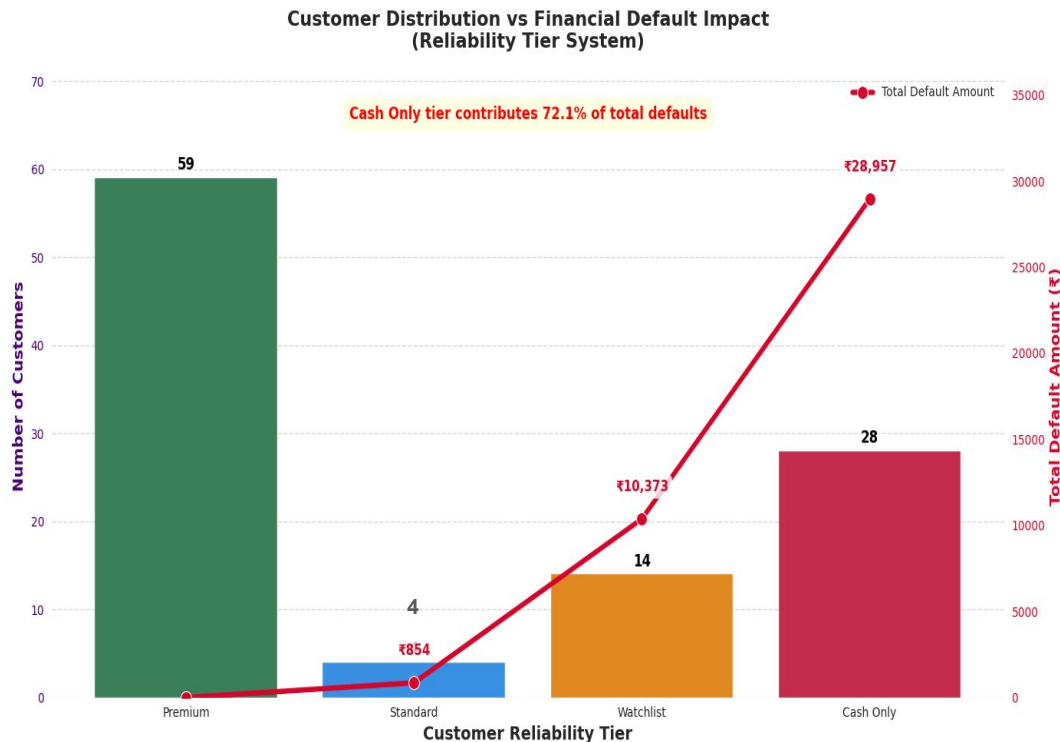
A single operational risk is materially distorting an otherwise healthy profit structure.

Liquidity Crisis: Cash-to-Credit Ratio Dropped to 'Warning' Zone



At the current trajectory, cash constraints will begin to restrict inventory restocking and limit strategic decisions such as promotions and investment in high-margin items.

The Leak: 28 'Cash Only' Customers Drive 72% of Defaults.



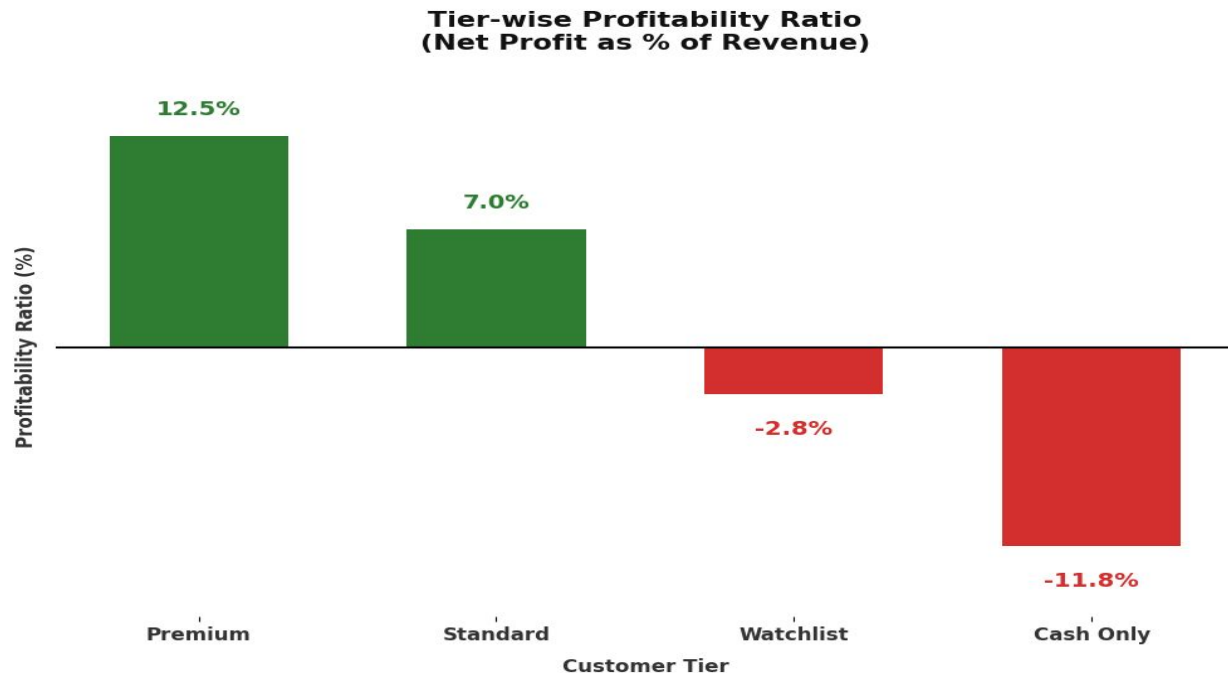
High Impact, Low Volume

- This tier contains only **28 customers**.
- Yet, it generates **₹28,957** in bad debt — **72% of total defaults**.

Recommendation: Immediate Credit Freeze

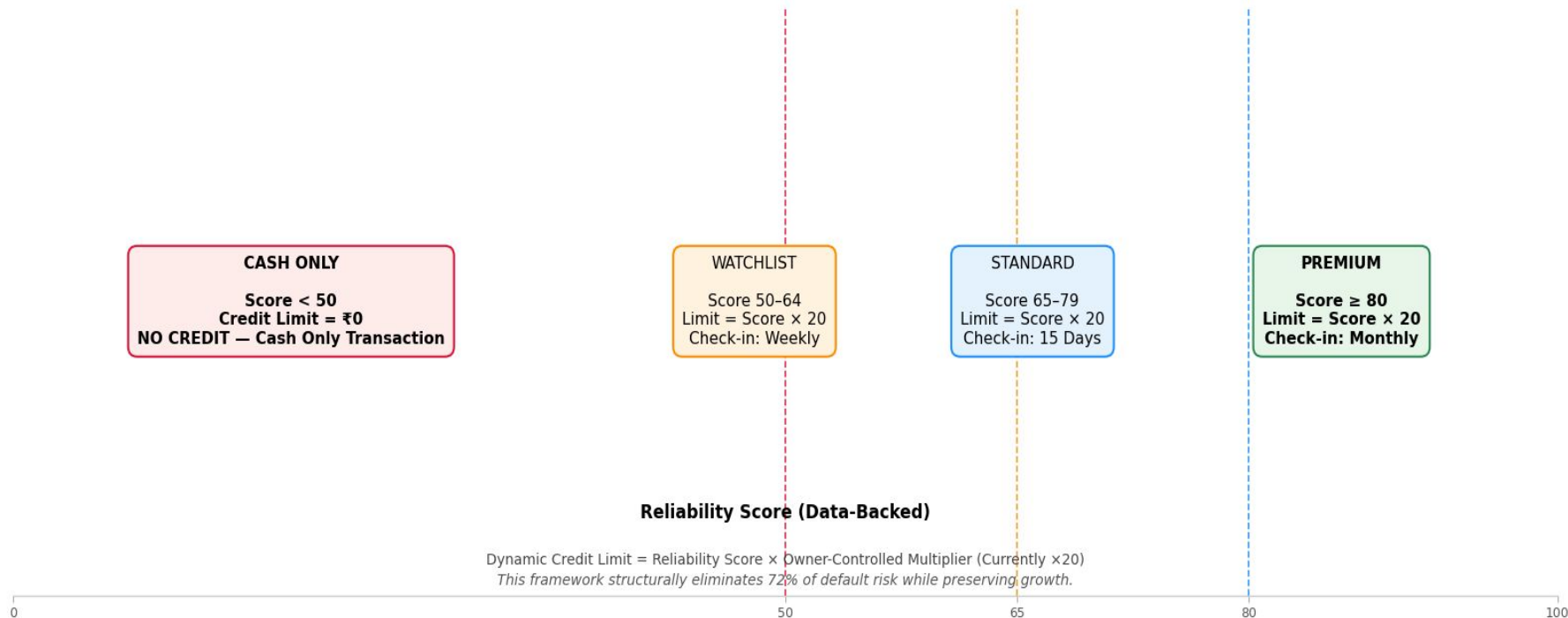
This action eliminates the single largest source of default risk.

Profit Impact: Servicing 'Cash Only' Customers Costs Us an 11.8% Net Loss

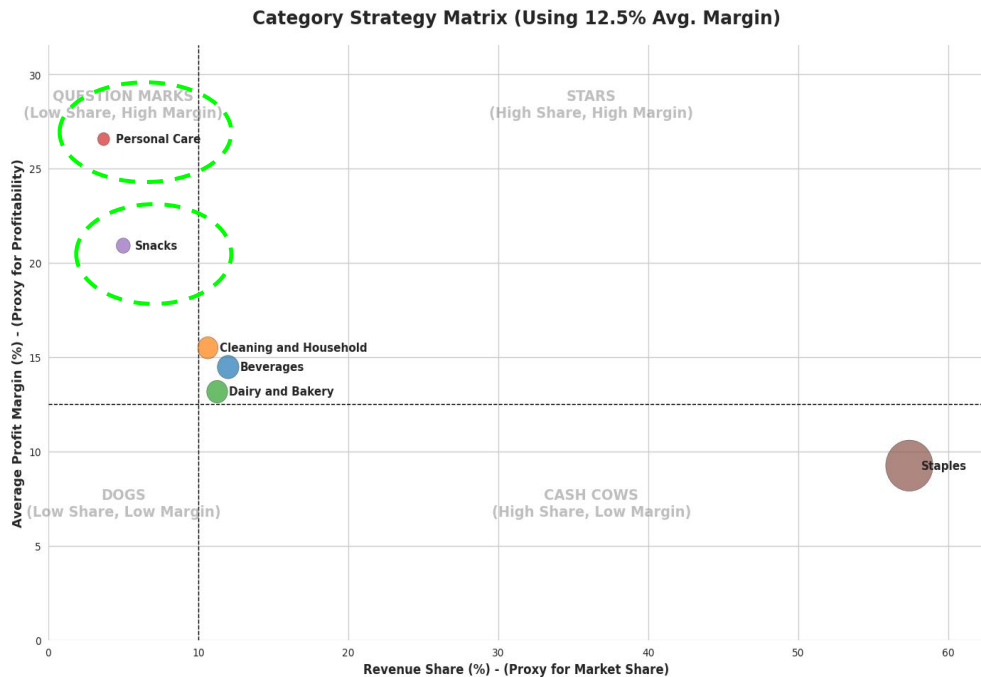


Not all revenue is good revenue. Every transaction with this segment destroys value.

Automating Credit Risk Control with Dynamic Limits



Growth Strategy: Shift Volume from Low-Margin Staples to High-Margin Personal Care & Snacks



The Trap

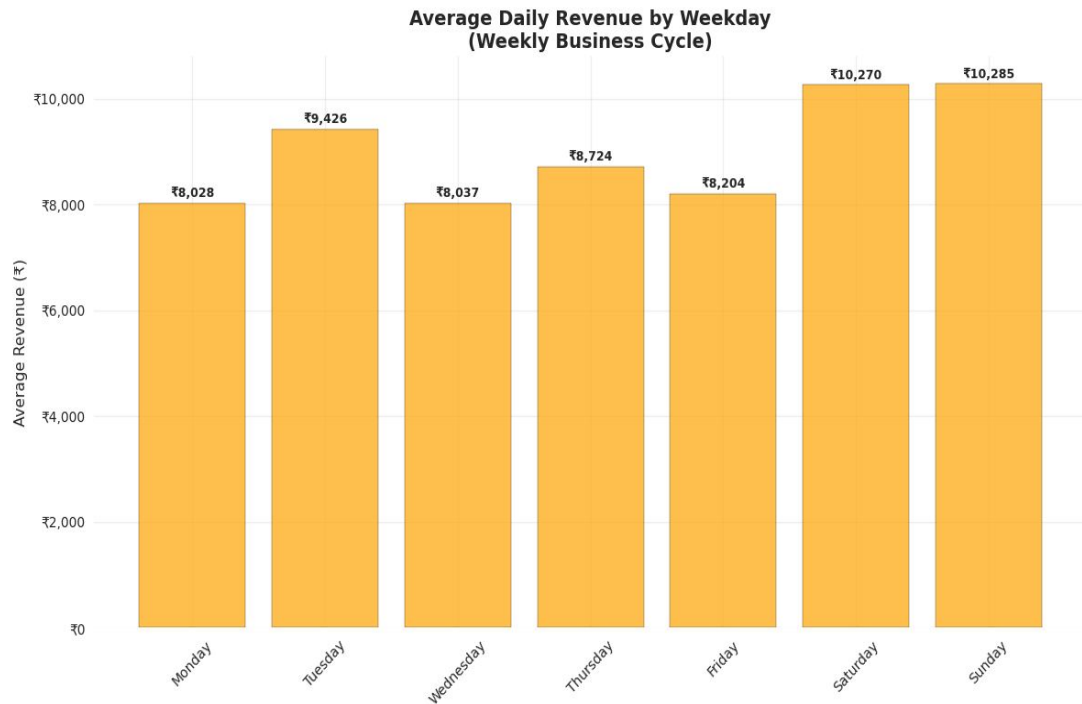
57% of revenue comes from Staples, delivering only 9.25% margin.

The Opportunity

Personal Care (26.6%) and Snacks (20.9%) deliver 2–3× higher margins, yet contribute less than 9% of revenue combined.

Margin expansion will come from category mix improvement, not forcing more volume.

Execution: Capitalizing on the +22% Weekend Traffic Surge



The Insight

Weekend revenue is ~22% higher than weekdays (hitting ₹10k+ daily).

Why it matters

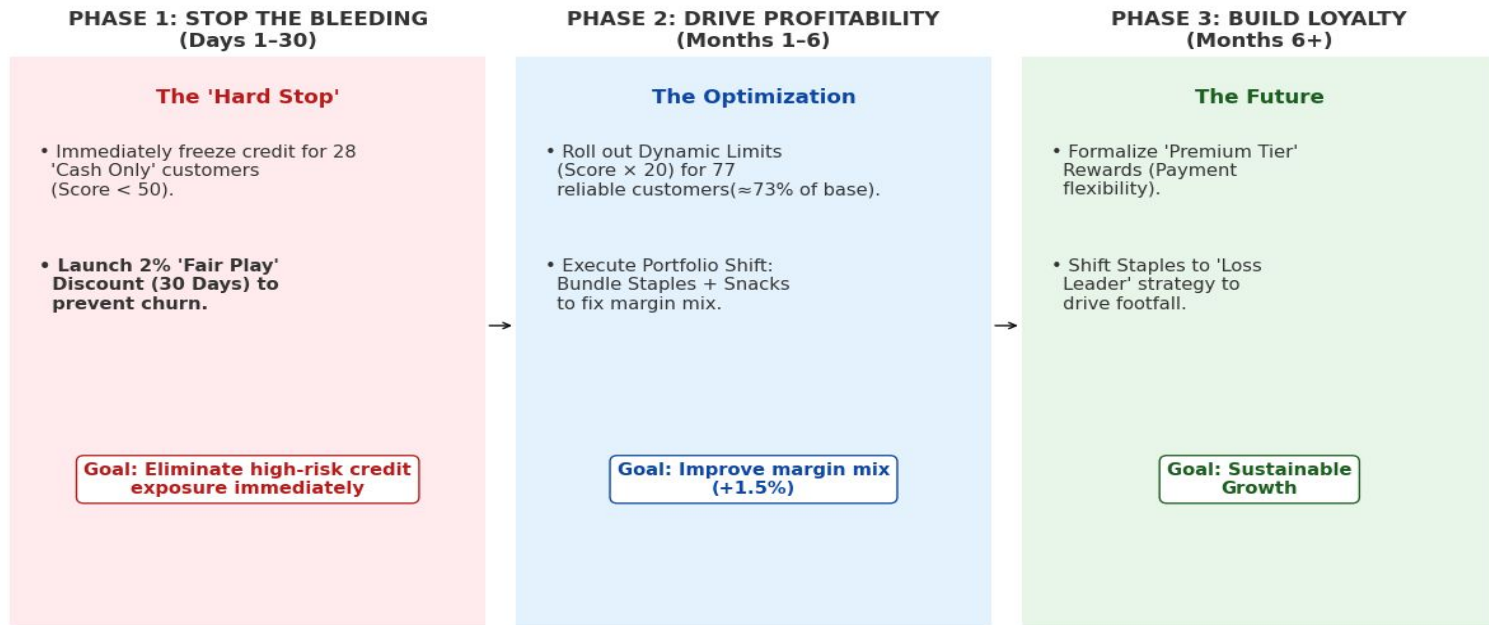
Traffic is already coming — **zero customer acquisition cost**.

The Action

Bundle and promote **high-margin Snacks & Personal Care** at the counter during weekends.

Growth is unlocked by timing + mix, not discounts.

Operational Roadmap & Risk Mitigation



Sequencing change to control risk before driving growth.

Projected Financial Impact: Financial Scorecard

VALUE CREATION LEVERS

1. Risk Mitigation

- Credit frozen for high-risk tiers (Cash Only + Watchlist)
- Dynamic limits for reliable customers

2. Margin Optimization

- Weekend bundling (Staples + Snacks)
- Mix shift toward high-margin SKUs

3. Operational Efficiency

- Improved cash-to-credit discipline
- Reduced bad-debt leakage

BUSINESS IMPACT (KPIs)

Net Profit Margin

9.45% → 13.50% **+405 bps**

Credit Risk Exposure

Concentrated → Controlled -85%

NET PROFIT UPLIFT: +43.4%

Source: 12-month financial summaries, 5-month granular credit audit, and 6-month category-wise daily summary data.

Appendix: Methodology, Data & Project Assets

VALUATION MODEL: PROFIT UPLIFT

(Basis: 5-Month Audited Financials)

1. BASELINE NET PROFIT ₹ 1,23,804
(Actuals @ 9.45% Margin)

2. RISK SAVINGS (Net) + ₹ 34,156
(Stop Credit for 72% Defaulters, 85% Eff.)

3. CATEGORY MIX UPLIFT + ₹ 19,642
(Personal Care/Snacks Shift, +1.5% Margin)

PROJECTED NET PROFIT ₹ 1,77,602

IMPACT SUMMARY:

- New Net Margin: 13.50% (+405 bps)
- Net Profit Uplift: +43.4%

CONSERVATIVE ASSUMPTIONS:

- Risk Efficiency: Model assumes 85% success in eliminating defaults via strict credit checks.
- Residual Risk: Remaining 15% default risk is conservatively excluded from the savings above.

```
def calculate_reliability_score(default_percentage, avg_delay):  
    default_percentage = float(default_percentage)  
    avg_delay = float(avg_delay)  
  
    # 70% weight on defaults, heavy penalty  
    default_score = 70 * max(0, (1 - (3 * default_percentage / 100)))  
  
    # 30% weight on payment delays (capped at 45 days)  
    delay_penalty = min(avg_delay / 45, 1)  
    delay_score = 30 * (1 - delay_penalty)  
  
    return max(0, min(100, default_score + delay_score))  
  
def assign_customer_tier(score):  
    if score >= 80: return "Premium"  
    elif score >= 65: return "Standard"  
    elif score >= 50: return "Watchlist"  
    else: return "Cash Only"
```

Project Assets

- ✓ Full Report (PDF)
- ✓ Google Colab Notebook (Code & Analysis)
- ✓ Dataset (Available, Client-confidential)